HSR-ACC-64 Optical Transport Failure Resolution SOP (Huawei Access Role)

Node: HSR-ACC-64 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_7534

Description

The HSR-ACC-64 access node is a critical component of the HSR Layout Access site, providing connectivity to the broader network infrastructure. This SOP addresses the specific scenario of Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Huawei-based access node, as indicated by the ALARM_7534 code with a Severity of Warning. The underlying protocol mechanics driving this failure domain stem from the Optical Transport Network (OTN) layer, which utilizes the generalized multiprotocol label switching (GMPLS) control plane to manage the transport of optical signals. The OTN layer is responsible for transporting client signals, such as SONET/SDH or Ethernet, over optical fibers. The GMPLS control plane enables the dynamic configuration of optical paths, including the allocation of SFP/XFP ports and the establishment of connections between nodes. However, this dynamic nature can sometimes lead to issues with SFP/XFP degradation or fiber cuts, resulting in Rx/Tx power loss and triggering the ALARM_7534 code. This SOP aims to guide the resolution of this specific failure domain, focusing on diagnostic, resolution, and verification steps tailored to the Huawei CLI dialect and the access role of the HSR-ACC-64 node.

Pre-requisite Checks

- Check: Verify the site-specific redundant link status to ensure a fallback path exists in case of a primary link failure.

- Check: Confirm the presence of remote hands at the HSR Layout Access site to facilitate physical access to the node.

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to optical signal attenuation or environmental factors	High
Fiber cuts or damage to the optical fibers connecting the node.	Medium
Rx/Tx power loss due to misconfigured or faulty optical connections.	Low

Diagnostic Commands & Expected Outputs

```
Command: display transceiver transceiver-id 1/0/1, Expected Output: Transceiver ID: 1/0/1, Description: This command isolates the specific SFP/XFP port under investigation, providing its ID and associated information, such as temperature and optical signal levels.

Command: display optical-power interface 1/0/1, Expected Output: Optical Power (dBm): 3.1, Description: This command verifies the Rx/Tx power levels on the specified interface, helping to identify potential
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power loss issues.

Command: display optical-fiber interface 1/0/1, Expected Output: Fiber Status: Normal, Description: This command checks the status of the optical fibers connected to the node, indicating whether there are any issues with fiber cuts or damage.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP port with a new one., Command: replace transceiver transceiver-id 1/0/1, Description: To resolve SFP/XFP degradation, replace the faulty SFP/XFP port with a new one, ensuring it is properly seated and secured.

Step 2: Step: Verify and adjust optical connections., Command: verify optical-connection interface 1/0/1, Description: To address Rx/Tx power loss due to misconfigured or faulty optical connections, verify and adjust the connections as necessary, ensuring proper alignment and secure mounting.

Step 3: Step: Repair or replace damaged optical fibers., Command: repair optical-fiber interface 1/0/1, Description: In the case of fiber cuts or damage, repair or replace the affected optical fibers, ensuring a secure and reliable connection between nodes.

Post-Check Verification

```
Command: display transceiver transceiver-id 1/0/1, Expected Output: Transceiver ID: 1/0/1 (active), Description: Verify that the replaced SFP/XFP port is now active and functioning correctly.

Command: display optical-power interface 1/0/1, Expected Output: Optical Power (dBm): 3.2, Description: Check that the Rx/Tx power levels have been restored to normal.

Command: display optical-fiber interface 1/0/1, Expected Output: Fiber Status: Normal, Description: Confirm that the optical fiber connections are secure and functioning correctly.
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Roll-back Procedures

- Step: Restore the original SFP/XFP port., Command: restore transceiver transceiver-id 1/0/1, Description: If the replacement SFP/XFP port fails to resolve the issue, restore the original port to its previous state.

- Step: Re-verify optical connections., Command: re-verify optical-connection interface 1/0/1, Description: Re-check the optical connections to ensure they are properly aligned and secured.

- Step: Repair or replace the original optical fibers., Command: re-repair optical-fiber interface 1/0/1, Description: If the repair or replacement of damaged optical fibers was attempted, re-attempt the repair or replacement as necessary.